

Juan Manuel Chau Delgado

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EDUCATION

- **University of Nottingham** October 2022 - September 2026
Psychology PhD Nottingham, United Kingdom
 - Thesis: "Exploring the factors and neural basis of visual working memory compression"
- **Pontificia Universidad Católica del Perú** March 2017 - December 2018
Mechatronics MEng Lima, Peru
 - Thesis: "Predefined Tasks Selection for an Assistive Robot for Disabled People through a Brain-Computer Interface using P300"
- **Universidad Nacional de Ingeniería** August 2007 - July 2013
Mechatronics BSc Lima, Peru
 - Thesis: "Motor Imagery-based Brain Signals Classification using Linear Discriminant Analysis with Regularization for People with Neurological Disability"

HONORS AND AWARDS

- **Brain Travel Grant** 2024
Brain: The Charity
 - £1000 awarded for conference travel expenses
- **Beca Generación del Bicentenario** 2022
PRONABEC Perú
 - Awarded to Peruvian citizens with top-level academic performance who lack the resources to fund graduate study programs
- **Beca Aristóteles** 2017
Pontificia Universidad Católica del Perú
 - Awarded to the overall top-ranked students from graduate programs

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=IN SUBMISSION

- [S.1] Abel Brodeur, Ghina Abdul Baki, Luna Fazio, et al. (2026). **Mass Reproducibility in Psychological Science**. Manuscript submitted for publication in *Psychological Science*. (Large-scale collaboration, ~200 co-authors, including **Juan M. Chau Delgado**).
- [S.2] Juan M. Chau Delgado, Joel Shebioba, Anna Pitt, Matias J. Ison, Nicholas E. Myers. (2026). **Chunking improves working memory via compression, not direct retrieval from long-term memory**. Manuscript submitted for publication in *Journal of Experimental Psychology: General*
- [J.1] **Juan M. Chau Delgado**, Matias J. Ison, Paul S. Muhle-Karbe, Mark G. Stokes, Sam Hall-McMaster, Nicholas E. Myers. (2025). **Alpha and Beta Oscillations Mediate the Effect of Motivation on Neural Coding of Cognitive Flexibility**. *Biological Psychology – Special Issue on Oscillations and Cognition*.
- [J.2] Rocio L. Callupe, **Juan M. Chau**, Jose G. Garcia, Christian G. Chicoma, Cesar F. Arrese, Manuel H. Artieda, Dante A. Elias, Elizabeth R. Villota. (2022). **Testing protocol for evaluating underhand serve-reception biomechanics in volleyball**. *Proceedings of the Institution of Mechanical Engineers, Part P: Journal of Sports Engineering and Technology*.
- [C.1] Victor Giron, **Juan M. Chau**, Anali Alfaro, Elizabeth R. Villota. (2021). **3D Human Pose Estimation for Martial Arts Analysis through Graph Convolutional Networks**. In *2021 14th International Conference on Machine Vision (ICMV)*.
- [C.2] Jorge Ramirez, **Juan M. Chau**. (2021). **Cognitive Performance Drop Detection During Daily Activities Using EEG**. In *2021 43rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*.
- [C.3] **Juan M. Chau**, Javier Hilario, Ever Lopez, Eduardo Bejar, Jorge Ramirez, Antonio Moran. (2021). **BINBOT: A Low-Cost Robot Companion that Teaches Basic Concepts through Binary Questions**. In *2021 7th International Conference on Control, Automation and Robotics (ICCAR)*.
- [J.3] **Juan M. Chau Delgado**, David Achancaray, Elizabeth R. Villota, Sylvain Chevallier. (2020). **Riemann-based Algorithms Assessment for Single and Multiple Trial P300 Classification in Non-optimal Environments**. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*.

- [C.4] David Achancaray, **Juan M. Chau**, Jairo Pirca, Francisco Sepulveda, Mitsuhiro Hayashibe. (2019). **Assistive Robot Arm Controlled by a P300-based Brain Machine Interface for Daily Activities**. In *2019 9th International IEEE/EMBS Conference on Neural Engineering (NER)*.
- [C.5] Amir Jahangiri, **Juan M. Chau**, David Achancaray, Francisco Sepulveda. (2018). **Covert Speech vs. Motor Imagery: a comparative study of class separability in identical environments**. In *2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*.
- [C.6] Gabriel Salvador, **Juan M. Chau**, Jorge Quesada, Cesar Carranza. (2018). **Efficient GPU-based implementation of the median filter based on a multi-pixel-per-thread framework**. In *2018 IEEE Southwest Symposium on Image Analysis and Interpretation (SSIAI)*.

RESEARCH EXPERIENCE

- **University of Nottingham** 2026
Research Assistant Nottingham, United Kingdom
 - Preprocessed MEG data, and developed a visual foraging task under the supervision of Professor Matias Ison
- **University of Nottingham** 2023
Research Assistant Nottingham, United Kingdom
 - Cleaned and restructured a decade's worth of experimental data sets from the University of Nottingham's Summer Scientist Week, under the supervision of Professor Lucy Cragg
- **University of Nottingham** 2021 - 2022
Independent Researcher Nottingham, United Kingdom
 - Studied cognitive processes and processing methods related to event-related potentials and reward processing, under the supervision of Professor Nicholas Myers
- **Pontificia Universidad Católica del Perú** 2019 - 2022
Research Fellow Lima, Peru
 - Studied volleyball and martial arts biomechanics to identify opportunities for improvement in physical training under the supervision of Professor Elizabeth Villota
- **Pontificia Universidad Católica del Perú** 2016 - 2019
Research Assistant Lima, Peru
 - Developed a non-invasive P300-based Brain-Computer Interface capable of performing remote control of an assistive robot using a EEG recordings under the supervision of Professor Elizabeth Villota
- **University of Essex** 2018
Research Intern Essex, United Kingdom
 - Designed and developed a non-invasive EEG acquisition experiment for a covert speech vs. motor imagery classification study under the supervision of Professor Francisco Sepulveda

ACADEMIC ADVISING

- **Pontificia Universidad Católica del Perú** 2019 - 2025
Undergraduate Thesis Supervisor Lima, Peru
 - Supervised 15 undergraduate senior theses (13 in Mechatronics, 2 in Biomedical Engineering), primarily on brain-computer interface applications (assistive robotics, rehabilitation devices)
- **Pontificia Universidad Católica del Perú** Lima, Peru
Undergraduate Thesis Committee Member
 - Served on thesis committees for 4 additional undergraduate students

TEACHING EXPERIENCE

- **University of Nottingham** 2022 - 2026
Demonstrator, School of Psychology Nottingham, United Kingdom
 - First Year Psychology Seminars, Practical Methods (Years 1, 2, and masters)
- **Pontificia Universidad Católica del Perú** 2019 - 2025
Lecturer, Department of Engineering Lima, Peru
 - Professional Development for Bioengineering, Thesis I & II (Biomedical Engineering), Thesis I & II (Mechatronics), Algorithms
- **Pontificia Universidad Católica del Perú** 2016 - 2022
Teaching Assistant, Department of Engineering Lima, Peru
 - Series and Transforms, Digital Signal and Image Processing, Digital Systems, Digital Signal Processing, Object-oriented Programming, Algorithms, Numerical Methods for Engineering, Automatic Control, Mechatronic Design Methodology

INDUSTRY EXPERIENCE

- **Bamtang Games S.A.C.**

Junior Developer

2013 - 2016

Lima, Peru

- Developed video games, video game engines and interactive tools using Haxe, C#, and JavaScript

SKILLS

- **Programming:** Proficient: MATLAB, Python. Familiar: Julia, R, C#, C/C++, JavaScript, Haxe, ActionScript 3, LabVIEW, Ladder, PHP
- **Research Tools & Software:** PsychoPy, signal processing software (MNE, Fieldtrip)
- **Languages:** Spanish (native), English (fluent), German (basic)
- **Additional training:** Neuromatch Academy: Computational Neuroscience (2023)